does not make it any less present, and, moreover, as the standard approach is vague regarding the quantum measurement problem—the very place where nonlocality makes itself apparent—once that issue is fully resolved, the nonlocality may not be so hidden after all. Others have argued that there are obstacles to making a relativistic version of the Bohm approach, although progress has been made on this front as well (see, for example, John Bell Beables *for Quantum Field Theory*, in the collected volume indicated above). And so, it is definitely worth keeping this alternative approach in mind, even if only as a foil against rash conclusions about what quantum mechanics unavoidably implies. For the mathematically inclined reader, a very nice treatment of Bohm's theory and issues of quantum entanglement can be found in Tim Maudlin, *Quantum Nonlocalityand Relativity* (Malden, Mass.: Blackwell, 2002).

14. For an in-depth, though technical, discussion of time's arrow in general, and the role of decoherence in particular, see H. D. Zeh, *The Physical Basis of the Direction of Time* (Heidelberg: Springer, 2001).

15. Just to give you a sense of how quickly decoherence takes place—how quickly environmental influence suppresses quantum interference and thereby turns quantum probabilities into familiar classical ones—here are a few examples. The numbers are approximate, but the point they convey is clear. The wavefunction of a grain of dust floating in your living room, bombarded by jittering air molecules, will decohere in about a billionth of a billionth of a billionth of a billionth (10^{-36}) of a second. If the grain of dust is kept in a perfect vacuum chamber and subject only to interactions with sunlight, its wavefunction will decohere a bit more slowly, taking a thousandth of a billionth of a billionth (10^{-21}) of a second. And if the grain of dust is floating in the darkest depths of empty space and subject only to interactions with the relic microwave photons from the big bang, its wavefunction will decohere in about a millionth of a second. These numbers are extremely small, which shows that decoherence for something even as tiny as a grain of dust happens very quickly. For larger objects, decoherence happens faster still. It is no wonder that, even though ours is a quantum universe, the world around us looks like it does. (See, for example, E. Joos, "Elements of Environmental Decoherence," in *Decoherence:Theoretical,*